

SAARTHI AI 2.0

Inclusive AI-Powered Government Scheme Recommendation Platform

Hack4Humanity Solution Documentation

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Abstract

Government welfare schemes provide financial assistance, healthcare benefits, educational support, employment opportunities, and social security to millions of citizens. However, identifying suitable schemes remains a significant challenge due to complex eligibility criteria, fragmented information, language barriers, and limited accessibility for differently abled users.

Saarthi AI 2.0 is an inclusive AI-powered Government Scheme Recommendation Platform designed to simplify scheme discovery through an accessibility-first approach. The platform enables users to receive personalized government scheme recommendations based on their profile while providing simplified explanations through an intuitive interface.

The solution is architected as:

React Frontend



Node.js + Express API



FastAPI AI Service



Supabase Database

The current hackathon MVP demonstrates the accessibility and recommendation workflow through the deployed frontend, while the backend services are designed as modular components for scalable future deployment.

The platform integrates accessibility features such as voice assistance, high-contrast interface, vibration feedback, and responsive design to ensure an inclusive experience for visually impaired users, hearing-impaired users, and citizens with low digital literacy. Saarthi AI 2.0 demonstrates how Artificial Intelligence and inclusive web technologies can improve citizen access to government welfare services while promoting digital inclusion.

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1. Introduction

Digital governance has transformed the way citizens access government welfare schemes and public services. Despite the availability of numerous central and state government schemes, many eligible beneficiaries remain unaware of the opportunities available to them due to fragmented information, complex eligibility criteria, language barriers, and limited digital literacy. These challenges are further intensified for visually impaired users, hearing-impaired users, senior citizens, and people living in rural areas with limited access to technology.

Saarthi AI 2.0 is an inclusive AI-powered Government Scheme Recommendation Platform developed to address these challenges through an accessibility-first approach. The platform enables users to discover government schemes based on their personal information while providing a simple, user-friendly, and accessible experience. By integrating Artificial Intelligence with modern web technologies, Saarthi AI 2.0 aims to simplify the process of scheme discovery and improve digital inclusion for all citizens.

2. Problem Statement

Government welfare schemes are designed to improve the quality of life of citizens by providing financial assistance, educational support, healthcare benefits, employment opportunities, housing schemes, and social security. However, a large number of eligible beneficiaries fail to access these schemes due to several challenges, including:

- I. Complex eligibility criteria that are difficult to understand.
- II. Information scattered across multiple government websites.
- III. Language barriers that make official content difficult to interpret.
- IV. Limited accessibility for visually impaired and hearing-impaired users.
- V. Low digital literacy among rural and elderly citizens.
- VI. Lack of personalized guidance for identifying relevant schemes.

These issues reduce awareness and prevent deserving citizens from receiving government benefits.

3. Existing System

The current process of finding suitable government schemes is primarily manual and time-consuming. Citizens are required to visit multiple government websites, read lengthy eligibility documents, and compare various schemes independently. Most existing portals provide limited personalization and lack accessibility features, making them difficult to use for people with disabilities or limited digital experience.

Limitations of the Existing System

- Manual search across multiple government portals.
 - Complex and lengthy eligibility criteria.
 - Limited accessibility support for differently abled users.
 - Absence of personalized recommendations.
 - Poor user experience for low-literacy and rural populations.
- Time-consuming process of comparing multiple schemes.

4. Proposed System

Saarthi AI 2.0 is an accessibility-first AI-powered Government Scheme Recommendation Platform that simplifies the process of identifying eligible government schemes. Users enter basic details such as age, income, occupation, state, and category. The platform analyzes the provided information and recommends relevant government schemes along with simplified explanations that are easy to understand.

The solution is architected as
React Frontend



Node.js + Express API



FastAPI AI Service



Supabase Database

The current hackathon MVP demonstrates the accessibility and recommendation workflow through the deployed frontend, while the backend services are designed as modular components for scalable future deployment.

The platform also incorporates accessibility features such as voice assistance, high-contrast interface, vibration feedback, and responsive design to ensure that users with different abilities can interact with the system independently.

5. Objectives

Primary Objectives

- Simplify the process of discovering government welfare schemes.
- Provide personalized recommendations based on user information.
- Improve accessibility for visually impaired, hearing-impaired, and low-literacy users.
- Reduce language and information barriers.
- Promote digital inclusion through an easy-to-use platform.

Secondary Objectives

- Support multilingual communication in future versions.
- Enable access in low-connectivity and rural environments.
- Develop a scalable architecture for future government API integration.
- Enhance transparency and awareness of government welfare programs.
- Encourage independent access to public services without intermediaries.

6. Target Users

Saarthi AI 2.0 is designed to serve a wide range of citizens who face challenges in accessing government welfare schemes. The primary target users are:

6.1 Visually Impaired Users

Citizens with visual impairments often find it difficult to navigate websites and read detailed government scheme information. Saarthi AI 2.0 provides voice assistance, screen-reader-friendly navigation, and a high-contrast interface to improve accessibility.

6.2 Hearing-Impaired Users

Hearing-impaired users may miss audio notifications or spoken guidance. The platform supports vibration feedback and visual alerts to ensure they receive important information effectively.

6.3 Low-Literacy Citizens

Many users struggle to understand complex government terminology and lengthy eligibility criteria. Saarthi AI 2.0 presents information in simple language with an easy-to-follow interface, making the platform more accessible.

6.4 Rural Citizens

People living in rural areas often have limited internet connectivity and lower digital literacy. The platform is designed with a simple workflow and is planned to support offline capabilities in future versions.

6.5 Students and Job Seekers

Students and job seekers frequently search for scholarships, skill development programs, and employment-related government schemes. Saarthi AI 2.0 helps them discover relevant schemes based on their eligibility.

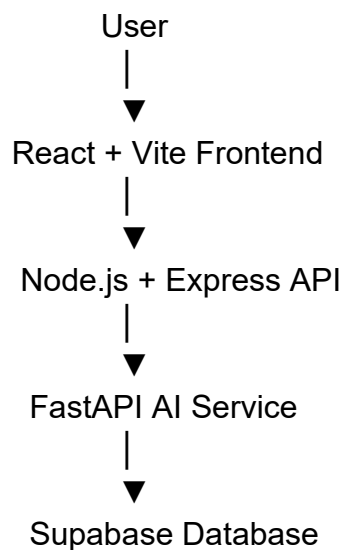
6.6 General Citizens

Any citizen looking for government welfare schemes can use the platform to receive personalized recommendations quickly and easily, without manually searching multiple government websites.

7. System Architecture

Saarthi AI 2.0 follows a modular architecture designed to provide scalability, maintainability, and accessibility. The system separates the frontend, backend, AI services, and database into independent components, making future enhancements easier to implement.

The proposed architecture is:



Component Description

React + Vite Frontend

The frontend provides a responsive and accessibility-focused user interface. It collects user information, displays recommendations, and includes accessibility features such as voice assistance, high-contrast mode, and responsive layouts.

Node.js + Express API

The API layer is designed to act as the communication bridge between the frontend and AI services. It manages request routing, validates user input, and coordinates communication between different system components.

FastAPI AI Service

The AI service is responsible for processing user information and applying eligibility rules to generate personalized government scheme recommendations. It is designed to support future AI enhancements and advanced recommendation techniques.

Supabase Database

Supabase serves as the cloud database for storing government scheme information and other application data. It is designed to provide secure, scalable, and efficient data management.

Note: The solution is architected as React → Node.js/Express → FastAPI → Supabase. The current hackathon MVP demonstrates the accessibility and recommendation workflow through the deployed frontend, while the backend services are designed as modular components for scalable future deployment.

8. Technology Stack

The development of Saarthi AI 2.0 combines modern web technologies, Artificial Intelligence, and cloud-based services to deliver an efficient and accessible government scheme recommendation platform.

Frontend

- [React.js](#)
- Vite

- HTML5
- CSS3
- JavaScript

Backend

- [Node.js](#)
- [Express.js](#)
- FastAPI

Database

- Supabase
- Artificial Intelligence
- AI-based Recommendation Engine
- Eligibility Evaluation Logic

Accessibility Technologies

- Web Speech API (Text-to-Speech)
- Vibration API
- High-Contrast Interface
- Screen Reader Support

Development Tools

- Git
- GitHub
- Visual Studio Code

Deployment Platform

- Vercel

9. Working Methodology

Saarthi AI 2.0 follows a structured workflow that simplifies government scheme discovery while ensuring accessibility for all users.

Step 1 – User Registration / Information Collection

The user enters basic personal information such as:

- Age
- Gender
- State
- Occupation
- Annual Income
- Social Category
- Education (if required)

Step 2 – Data Validation

The application verifies that all required information has been entered correctly before proceeding with the recommendation process.

Step 3 – Eligibility Evaluation

The AI recommendation engine analyzes the user's information and applies predefined eligibility rules to identify government schemes that match the user's profile.

Step 4 – Recommendation Generation

Based on the evaluation results, the system generates personalized government scheme recommendations along with simplified explanations that help users understand why each scheme is relevant.

Step 5 – Accessibility Support

To ensure inclusive access, the platform provides accessibility features such as:

- Voice read-aloud assistance
- High-contrast interface

Screen-reader compatibility
Vibration feedback on supported devices

Step 6 – Recommendation Display

The recommended government schemes are presented through a clean and user-friendly interface, allowing users to explore relevant schemes easily.

10. Implemented Features

The current Hack4Humanity MVP of Saarthi AI 2.0 includes several accessibility-focused and AI-assisted features.

AI-Based Recommendation System

Provides personalized government scheme recommendations based on user information.

Voice Read-Aloud Assistance

Uses Text-to-Speech technology to read recommendations aloud for visually impaired users.

High-Contrast Interface

Improves readability and visibility for users with low vision.

Responsive User Interface

Ensures compatibility across desktop, tablet, and mobile devices.

Simplified Recommendation Explanations

Displays recommendations using clear and easy-to-understand language.

Accessibility-First Design

The interface is designed to be usable by people with different abilities and varying levels of digital literacy.

Vibration Feedback

Provides tactile feedback on supported devices to improve interaction for hearing-impaired users.

11. Accessibility Features

Accessibility is the core principle behind Saarthi AI 2.0. The platform aims to provide equal access to government welfare information for all users.

Accessibility for Visually Impaired Users

- Voice read-aloud assistance
- Screen-reader-friendly interface
- High-contrast display mode
- Clear navigation structure

Accessibility for Hearing-Impaired Users

- Vibration feedback
- Visual notifications
- Text-based interaction
- Clear visual indicators

Accessibility for Low-Literacy Users

Simple and easy-to-understand language
Clean user interface
Guided interaction
Reduced technical complexity

Accessibility for Rural Users

Responsive interface
Lightweight web application
Future support for offline usage
Simplified navigation for first-time users.

Inclusive Design Principles

The platform follows an accessibility-first approach by ensuring that users with diverse abilities can independently discover and understand government welfare schemes without relying on intermediaries.

12. AI Recommendation Workflow

The recommendation process in Saarthi AI 2.0 is designed to provide users with personalized government scheme suggestions through a simple and accessible workflow. The system analyzes the user's information, evaluates eligibility conditions, and presents suitable schemes in an easy-to-understand format.

Step 1: User Profile Collection

The platform collects essential user details required for eligibility evaluation, such as:

- Age
- State
- Annual Income
- Occupation
- Social Category
- Educational Qualification (where applicable)

These details are used to identify government schemes that best match the user's profile.

Step 2: Eligibility Evaluation

The collected information is processed using predefined eligibility rules. The system compares the user's profile with the eligibility criteria of various government schemes to determine suitable matches.

Step 3: AI-Based Recommendation

Based on the eligibility evaluation, the recommendation engine identifies the most relevant government schemes and generates personalized recommendations.

The AI layer is designed to simplify the recommendation process by providing relevant and user-friendly results instead of requiring users to manually search multiple government portals.

Step 4: Simplified Explanation

For every recommended scheme, the platform provides a simple explanation describing why the user may be eligible. This helps users understand the recommendation without reading lengthy government documents.

Future AI Enhancements

The architecture is designed to support future AI capabilities, including:

- Vector-based semantic search using pgvector
- LLM-assisted eligibility verification Confidence scoring for recommendations
- Intelligent ranking of schemes
- Context-aware recommendation improvements.

13. Implementation Details

Saarthi AI 2.0 has been developed using modern web technologies following a modular architecture. Each component of the system has a specific responsibility, making the application scalable and easier to maintain.

Frontend Implementation

The frontend is developed using React.js with Vite, providing a responsive and interactive user interface. It manages user interactions, collects profile information, and displays personalized recommendations while incorporating accessibility features.

Backend Architecture

The solution is architected using Node.js, Express, and FastAPI to support modular backend services. The backend architecture is designed to validate requests, process eligibility information, and coordinate communication between different system components.

The current hackathon MVP primarily demonstrates the frontend workflow, while the backend services are designed for future deployment and scalability.

Database Design

Supabase is selected as the cloud database solution due to its scalability, security, and efficient data management capabilities. It is intended to manage government scheme information and support future application requirements.

Accessibility Implementation

Accessibility is integrated throughout the platform rather than being treated as an additional feature. Technologies such as Text-to-Speech, high-contrast themes, responsive layouts, and vibration feedback ensure that users with different abilities can interact with the platform comfortably.

14. Deployment

The Saarthi AI 2.0 project follows a modern deployment strategy suitable for scalable web applications.

GitHub Repository

The project source code is maintained using GitHub, enabling version control, collaborative development, and efficient project management.

<https://github.com/suhani1801/Saarthi-AI-2.0>

Live demo

<https://saarthi-ai-2-0.vercel.app>

Frontend Deployment

The accessibility-focused frontend is deployed using Vercel, allowing users to access the application through a web browser.

Backend Deployment

The backend architecture has been designed as modular services using Node.js, Express, FastAPI, and Supabase. While the current hackathon MVP demonstrates the frontend experience, the backend architecture supports future deployment and integration.

15. Project Impact

Saarthi AI 2.0 aims to create both social and technical impact by making government welfare schemes more accessible to citizens.

Social Impact

- Improves awareness of government welfare schemes.
- Encourages equal access to public services.
- Supports visually impaired and hearing-impaired users.
- Helps low-literacy citizens understand government benefits.
- Promotes digital inclusion across different communities.

Technical Impact

- Demonstrates the practical use of Artificial Intelligence in public services.
- Implements accessibility-first web development practices.
- Utilizes a scalable and modular software architecture.
- Encourages future integration with government digital platforms.

16. Advantages of Saarthi AI 2.0

The platform offers several advantages compared to traditional government scheme search methods.

- Personalized government scheme recommendations.
- Accessibility-focused interface for users with disabilities.
- Simplified explanations that improve user understanding.
- Modular architecture for future expansion. AI-assisted recommendation workflow.
- Easy-to-use interface requiring minimal technical knowledge.
- Responsive design supporting multiple devices.
- Scalable architecture suitable for future enhancements.
- Reduced time required to identify relevant government schemes.
- Improved user experience through inclusive design principles.

17. Future Scope

Saarathi AI 2.0 has significant potential for future enhancements to improve its functionality and user experience. Some proposed improvements include:

- Speech-to-Text form filling for hands-free interaction.
- OCR-based document extraction from ID cards and certificates.
- Multilingual support for regional languages.
- Offline support for users in low-connectivity areas.
- Real-time government scheme updates.
- Administrative dashboard for scheme management.
- Advanced AI features such as semantic search and confidence-based recommendations.

These enhancements will make the platform more scalable, intelligent, and accessible to a larger number of users.

18. Conclusion

Saarathi AI 2.0 is an accessibility-first AI-powered Government Scheme Recommendation Platform that helps citizens easily discover government welfare schemes based on their eligibility. The platform combines Artificial Intelligence with modern web technologies to provide a simple, inclusive, and user-friendly experience.

With features such as voice assistance, high-contrast mode, responsive design, and personalized recommendations, Saarthi AI 2.0 aims to improve digital accessibility and promote equal access to government services. Its modular architecture also provides a strong foundation for future enhancements, making it a scalable and socially impactful solution.

19. References

The following resources were referred to during the development of Saarthi AI 2.0:

1. Government of India - Official Government Schemes and Services
2. MyScheme Portal
3. React Official Documentation
4. Vite Official Documentation
5. Node.js Official Documentation
6. Express.js Official Documentation
7. FastAPI Official Documentation
8. Supabase Official Documentation
9. MDN Web Docs Web Speech API
10. GitHub Documentation
11. Vercel Documentation